15

Industrial Ethernet

Objectives

When you have completed study of this chapter, you will be able to:

- Describe the concept of Industrial Ethernet with specific reference to:
  - Connectors and cabling
  - Packaging
  - Determinism
  - Power on the bus
  - Redundancy

15.1 Introduction

Early Ethernet was not entirely suitable for control functions as it was primarily developed for office–type environments. The Ethernet technology has, however, made rapid advances over the past few years. It has gained such widespread acceptance in Industry that it is becoming the *de facto* field bus technology. An indication of this trend is the inclusion of Ethernet as the levels 1 and 2 infrastructure for Modbus/TCP (Schneider), Ethernet/IP (Rockwell Automation and ODVA), ProfiNet (Profibus) and Foundation Fieldbus HSE.

The following sections will deal with problems related to early Ethernet, and how they have been addressed in subsequent upgrades.

15.2 Connectors and cabling

Earlier industrial Ethernet systems such as the first–generation Siemens SimaticNet (Sinec–H1) were based on the 10Base5 configuration, and thus the connectors involved include the screw–type N–connectors and the D–type connectors, which are both fairly rugged. The heavy–gauge twin–screen (braided) RG–8 coaxial cable is also quite impervious to electrostatic interference.

Most modern industrial Ethernet systems are, however, based on a 10BaseT/100BaseTX configuration and thus have to contend with RJ–45 connectors and Cat5-type UTP cable. The RJ-45 connectors could be problematic. They are everything

but rugged and are suspect when subjected to great temperature extremes, contact with oils and other fluids, dirt, UV radiation, EMI as well as shock, vibration and mechanical loading.

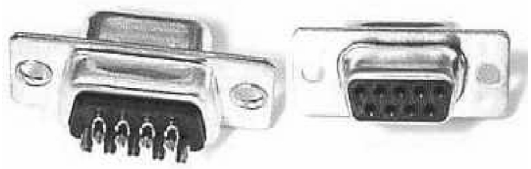


Figure 15.1
D-type connectors

As in interim measure, some manufacturers have started using D–type (known also as DB or D–Subminiature) connectors. These are mechanically quite rugged, but are neither waterproof nor dustproof either. They can therefore be used only in IP20 rated environments, i.e. within enclosures in a plant.

Ethernet I/O devices have become part of modern control systems. Ethernet makes it possible to use a variety of TCP/IP protocols to communicate with decentralized components virtually to sensor level. As a result, Ethernet is now installed in areas that were always the domain of traditional Fieldbus systems. These areas demand IP67 class protection against dirt, dust and fluids. This requires that suitable connector technology meeting IP67 standards have to be defined for transmission speeds up to 100Mbps. Two solutions to this problem are emerging. The one is a modified RJ-45 connector while the other is an M12 (micro-style) connector.



Figure 15.2
Modified RJ-45 connector(RJ-LNxx)
Courtesy: AMC Inc

Standardization groups are addressing the problem both nationally and internationally. User organizations such as IAONA (Industrial Automation Open Networking Alliance), Profibus user organization and ODVA (Open DeviceNet Vendor Association) are also trying to define standards within their organizations.

Network connectors for IP67 are not easy to implement. Three different approaches can be found. First, there is the RJ-45 connector sealed within an IP67 housing. Then there is an M12 (micro-style) connector with either four or eight pins. A third option is a hybrid connector based on RJ-45 technology with additional contacts for power distribution. Two of the so-called sealed RJ-45 connectors are in the process of standardization. Initially the 4-pin M12 version will be standardized in Europe. Connectors will be tested against existing standards (e.g., VDE 0110) and provide the corresponding IP67 class

protection at 100 Mbps. In the US, the ODVA has standardized a sealed version of the RJ-45 connector for use with Ethernet/IP.

For the use of the standard M12 in Ethernet systems is covered in standard EN 61076-2-101. The transmission performance of 4-pin M12 connector for Ethernet up to 100Mbps is comparable, if not better than standardized office grade Ethernet products. In office environments, four-pair UTP cabling is common. For industrial applications two-pair cables are less expensive and easier to handle. Apart from installation difficulties, 8-pin M12 connectors may not meet all the electrical requirements described in EN 50173 or EIA/TIA-568B.



Figure 15.3
M12 connector (EtherMate)
Courtesy: Lumberg Inc.

Typical M12 connectors for Ethernet are of the 8–pole variety, with threaded connectors. They can accept various types of Cat5/5e twisted pair wiring such as braided or shielded wire (solid or stranded), and offer excellent protection against moisture, dust, corrosion, EMI, RFI, mechanical vibration and shock, UV radiation, and extreme temperatures (–40ºC to 75ºC).

As far as the media is concerned, several manufacturers are producing Cat5 and Cat5e wiring systems using braided or shielded twisted pairs. An example an integrated approach to industrial Ethernet cabling is Lumberg's etherMATE™ system, which includes both the cabling and an M12 connector system.

Some vendors also use 2-pair Cat5+ cable, which has an outer diameter similar to Cat5 cable, but have wire with a thicker cross-section. This, together with special punch-down connectors, greatly simplifies installation.

15.3 Packaging

Commercial Ethernet equipment (hubs, switches, etc) are only rated to IP20, in other words, they have to be deployed in enclosures for industrial applications. They are also typically rated to 40 ^{o}C. Additional issues relate to vibration and power supplies.

Some manufacturers are now offering industrially-hardened switches with DIN-rail mounts, IP67 (waterproof and dustproof) rating, industrial temperature rating (60°C), DIN-rail mounts and redundant power supplies



Figure 15.4
Industrial grade switch
Courtesy: Siemens

15.4 Deterministic versus stochastic operation

One of the most common complaints with early Ethernet was that it uses CSMA/CD (a probabilistic or stochastic method) as opposed to other automation technologies such as Fieldbus that use deterministic access methods such as token passing or the publisher–subscriber model. CSMA/CD essentially means that it is impossible to guarantee delivery of a possibly critical message within a certain time. This could be due either to congestion on the network (often due to other less critical traffic) or to collisions with other frames. In office applications there is not much difference between 5 seconds and 500 milliseconds, but in industrial applications a millisecond counts. Industrial processes often require scans in a 5 to 20–millisecond range, and some demanding processes could even require 2 to 5 milliseconds. On 10BaseT Ethernet, for example, the access time on a moderately loaded 100–station network could range from 10 to 100mS, which is acceptable for office applications but not for processes.

There is a myth doing the rounds that Ethernet will experience an exponential growth in collisions and traffic delays- resulting ultimately in a collapse of the network- if loaded above 40%. In fact, Ethernet delays for 10 Mbps Ethernet are linear and can be consistently maintained under 2 ms for a lightly loaded network (<10%) and 30 ms for a heavily loaded network (<50%).

It is therefore important for the loading or traffic needs to be carefully analyzed to ensure that the network is not overwhelmed at critical or peak operational times. While a typical utilization factor on a commercial Ethernet LAN of 30% is acceptable, figures of less than 10% utilization on an industrial Ethernet LAN are required. Most industrial networks run at 3 or 4% utilization with fairly large number of I/O points being transferred across the system.

The advent of Fast and Gigabit Ethernet, switching hubs, IEEE 802.3Q VLAN technology, IEEE 802.3p traffic prioritization and full duplex operation has resulted in

very deterministic Ethernet operation and has effectively put this concern to rest for most applications.

15.5 Size and overhead of Ethernet frame

Data link encoding efficiency is another problem, with the Ethernet frames taking up far more space than for an equivalent fieldbus frame. If the TCP/IP protocol is used in addition to the Ethernet frame, the overhead increases dramatically. The efficiency of the overall system is, however, more complex than simply the number of bytes on the transmitting cable and issues such as raw speed on the cable and the overall traffic need to be examined carefully. For example, if 2 bytes of data from an instrument had to be packaged in a 60 byte message (because of TCP/IP and Ethernet protocols being used) this would result in an enormous overhead compared to a fieldbus protocol. However, if the communications link were running at 100 Mbps or 1 Gbps with full duplex communications, this would put a different light on the problem and make the overhead issue almost irrelevant.

15.6 Noise and interference

Due to higher electrical noise near to the industrial LANs some form of electrical shielding and protection is useful to minimize errors in the communication. A good choice of cable is fiber–optic (or sometimes coaxial cable). Twisted pair can be used but care should be taken to route the cables far away from any potential sources of noise. If twisted pair cable is selected; a good decision is to use screened twisted pair cable (ScTP) rather than the standard UTP.

It should be noted here that Ethernet–based networks are installed in a wide variety of systems and rarely have problems reported due to reliability issues. The use of fiber ensures that there are minimal problems due to earth loops or electrical noise and interference.

15.7 Partitioning of the network

It is very important that the industrial network operates separately from that of the commercial network, as speed of response and real time operation are often critical attributes of an industrial network. An office type network may not have the same response requirements. In addition, security is another concern where the industrial network is split off from the commercial networks so any problems in the commercial network will not affect the industrial side.

Industrial networks are also often partitioned into individual sub–networks for reasons of security and speed of response, by means of bridges and switches.

In order to reduce network traffic, some PLC manufacturers use exception reporting. This requires only changes in the various digital and analog parameters to be transmitted on the network. For example, if a digital point changes state (from ‘on’ to ‘off’ or ‘off’ to ‘on’), this change would be reported. Similarly, an analog value could have associated with it a specified change of span before reporting the new analog value to the master station.

15.8 Switching technology

Both the repeating hub and bridge technologies are being superseded by switching technology. This allows traffic between two nodes on the network to be directly

connected in a full duplex fashion. The nodes are connected through a switch with extremely low levels of latency. Furthermore, the switch is capable of handling all the ports communicating simultaneously with one another without any collisions. This means that the overall speed of the switch backplane is considerably greater than the sum of the speeds of the individual Ethernet ports.

Most switches operate at the data link layer and are also referred to as switching hubs and layer 2 switches. Some switches can interpret the network layer addresses (e.g. the IP address) and make routing decisions on that. These are known as layer 3 switches.

Advanced switches can be configured to support virtual LANs. This allows the user to configure a switch so that all the ports on the switch are subdivided into predefined groups. These groups of ports are referred to as virtual LANs (VLANs)– a concept that is very useful for industrial networks. Only switch ports allocated to the same VLAN can communicate with each other.

Switches do have performance limitations that could affect a critical industrial application. If there is traffic on a switch from multiple input ports aimed at one particular output port, the switch may drop some of the packets. Depending on the vendor implementation, it may force a collision back to the transmitting device so that the transmitting node backs off long enough for the congestion to clear. This means that the transmitting node does not have a guarantee on the transmission between two nodes – something that could impact a critical industrial application.

In addition, although switches do not create separate broadcast domains, each virtual LAN effectively forms one (if this is enabled on the switch). An Ethernet broadcast message received on one port is retransmitted onto all ports in the VLAN. Hence a switch will not eliminate the problem of excessive broadcast traffic that can cause severe performance degradation in the operation of the network. TCP/IP uses the Ethernet broadcast frame to obtain MAC addresses and hence broadcasts are fairly prevalent here.

A problem with a switched network is that duplicate paths between two given nodes could cause a frame to be passed around and around the 'ring' caused by the two alternative paths. This possibility is eliminated by the 'Spanning Tree' algorithm, the IEEE802.1d standard for layer 2 recovery. However, this method is quite slow and could take from 2 to 5 seconds to detect and bypass a path failure and could leave all networked devices isolated during the process. This is obviously unacceptable for industrial applications.

A solution to the problem is to connect the switches in a dual redundant ring topology, using copper or fiber. This poses a new problem, as an Ethernet broadcast message will be sent around the loop indefinitely. Several vendors now market switches with added redundancy management capabilities. One of the switches in the system acts as a redundancy manager and allows a physical 200 Mbps ring to be created, by terminating both ends of the traditional Ethernet bus in itself. Although the bus is now looped back to itself, the redundancy manager logically breaks the loop, preventing broadcast messages from wreaking havoc. Logically, the redundancy manager behaves like two nodes, sitting back to back, transmitting and receiving messages to the other around the ring using 802.1p/Q frames. This creates a deterministic path through any 802.1p/Q compliant switches (up to 50) in the ring, which results in a real time 'awareness' of the state of the ring. Up to 50 switches can be interconnected in this way, using 3 km fiber connections. As a result, a dual fiber-optic ring with a circumference of 150 km can be created!

When a network failure is detected (i.e. the loop is broken), the redundancy manager interconnects the two segments attached to it, thereby restoring the loop. This takes place in between 20 and 500 milliseconds, depending on the size of the ring.

15.9 Power on the bus

Industry often expects device power to be delivered over the same wires as those used for communicating with the devices. Examples of such systems are DeviceNet and Foundation Fieldbus. This is, however, not an absolute necessity as the power can be delivered separately. Profibus DP, for example, does not provide this feature yet it is one of the leading Fieldbuses.

Ethernet does, however, provide the ability to deliver some power. The IEEE 802.3af standard was ratified by the IEEE Standards Board in June 2003 and allows a source device (a hub or a switch) to supply a minimum of 300 mA at 48 Volts DC to the field device. This is in the same range as FF and DeviceNet.

The standard allows for two alternatives, namely the transmission of power over the signal pairs (1/2 and 3/6) or the transmission of power over the unused pairs (4/5 and 7/8). Intrinsic safety issues still need addressing.

15.10 Fast and Gigabit Ethernet

The recent developments in Ethernet technology are making it even more relevant in the industrial market. Fast Ethernet as defined in the IEEE specification 802.3u is essentially Ethernet running at 100 Mbps. The same frame structure, addressing scheme and CSMA/CD access method are used as with the 10 Mbps standard. In addition, Fast Ethernet can also operate in full duplex mode as opposed to CSMA/CD, which means that there are no collisions. Fast Ethernet operates at ten times the speed of that of standard IEEE 802.3 Ethernet. Video and audio applications can enjoy substantial improvements in performance using Fast Ethernet. Smart instruments that require far smaller frame sizes will not see such an improvement in performance. One area however, where there may be significant throughput improvements, is in the area of collision recovery. The back-off times for 100 Mbps Ethernet is a tenth that of standard Ethernet. Hence a heavily loaded network with considerable individual messages and nodes would see performance improvements. If loading and collisions are not really an issue on the slower 10 Mbps network, then there will not be many tangible improvements in the higher LAN speed of 100 Mbps.

Note that with the auto–negotiation feature built into standard switches and many Ethernet cards, the device can operate at either the 10 or 100 Mbps speeds. In addition, the Cat5 wiring installed for 10BaseT Ethernet is adequate for the 100 Mbps standard as well.

Gigabit Ethernet is another technology that could be used to connect instruments and PLCs. However its speed would probably not be fully exploited by the instruments for the reasons indicated above.

15.11 TCP/IP and industrial systems

The TCP/IP suite of protocols provides for a common open protocol. In combination with Ethernet this can be considered to be a truly open standard available to all users and vendors. However, there are some problems at the application layer area. Although TCP/IP implements four layers which are all open (network interface, internet, transport and application layers), most industrial vendors still implement their own specific application layer. Hence equipment from different vendors can coexists on the factory shop floor but cannot inter–operate. Protocols such as MMS (manufacturing messaging service) have been promoted as truly ‘open’ automation application layer protocols but with limited acceptance to date.

15.12 Industrial Ethernet architectures for high availability

There are several key technology areas involved in the design of Ethernet based industrial automation architecture. These include available switching technologies, quality of service (QoS) issues, the integration of existing (legacy) field buses, sensor bus integration, high availability and resiliency, security issues, long distance communication and network management– to name but a few.

For high availability systems a single network interface represents a single point of failure (SPOF) that can bring the system down. There are several approaches that can be used on their own or in combination, depending on the amount of resilience required (and hence the cost). The cost of the additional investment in the system has to be weighed against the costs of any downtime.

For a start, the network topology could be changed to represent a switched ring. Since the Ethernet architecture allows an array of switches but not a ring, this setup necessitates the use of a special controlling switch (redundancy manager) which controls the ring and protects the system against a single failure on the network. It does not, however, guard against a failure of the network interface on one of the network devices. The redundancy manager is basically a switch that "divides" itself in two internally, resulting in two halves that periodically check each other by sending high-priority messages around the loop to each other.

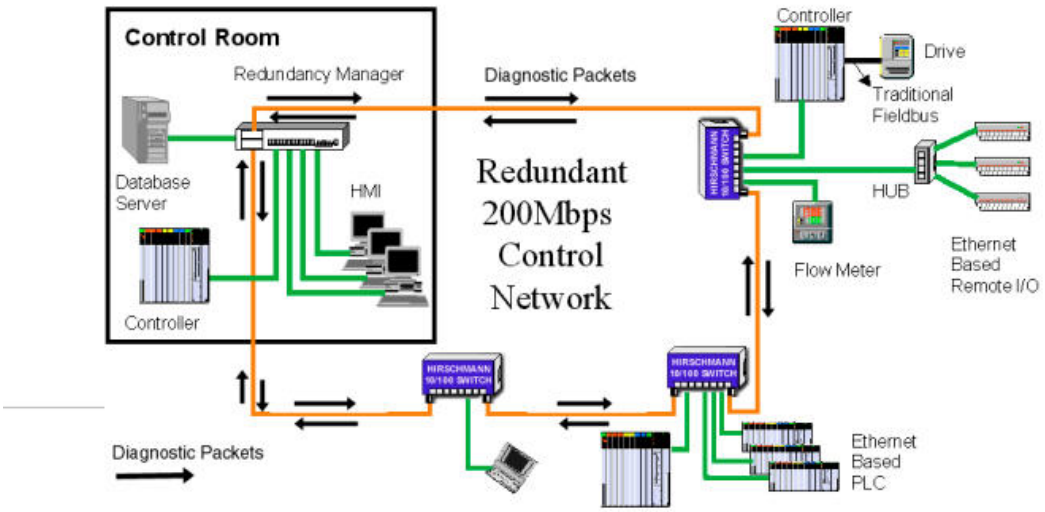


Figure 15.5
Redundant switch ring
Courtesy: Hirschmann

If a failure occurs anywhere on the ring, the redundancy manager becomes aware of it through a failure to communicate with itself, and "heals" itself.

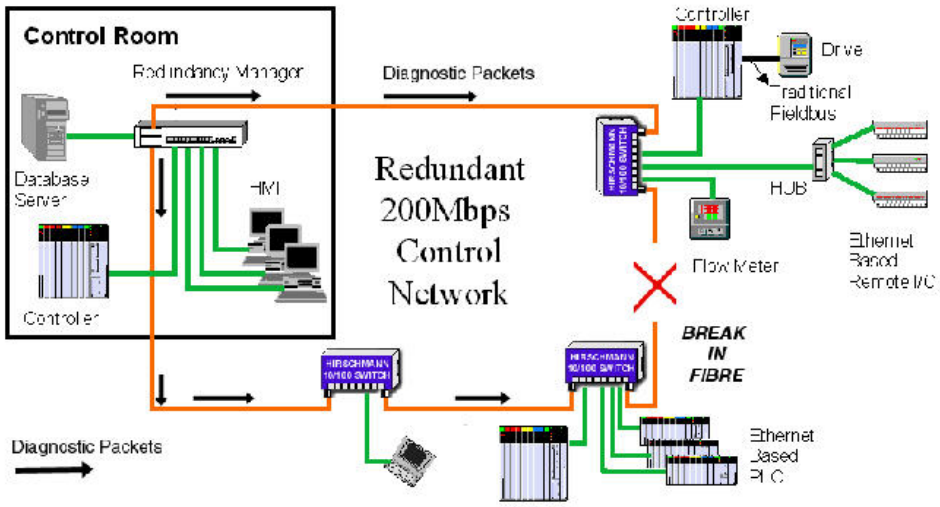


Figure 15.6
Redundant switch ring with failure
Courtesy: Hirschmann

The next level of resiliency would necessitate two network interfaces on the controller (that is, changing it to a dual–homed system), each one connected to a different switch. This setup would be able to tolerate both a single network failure and a network interface failure.

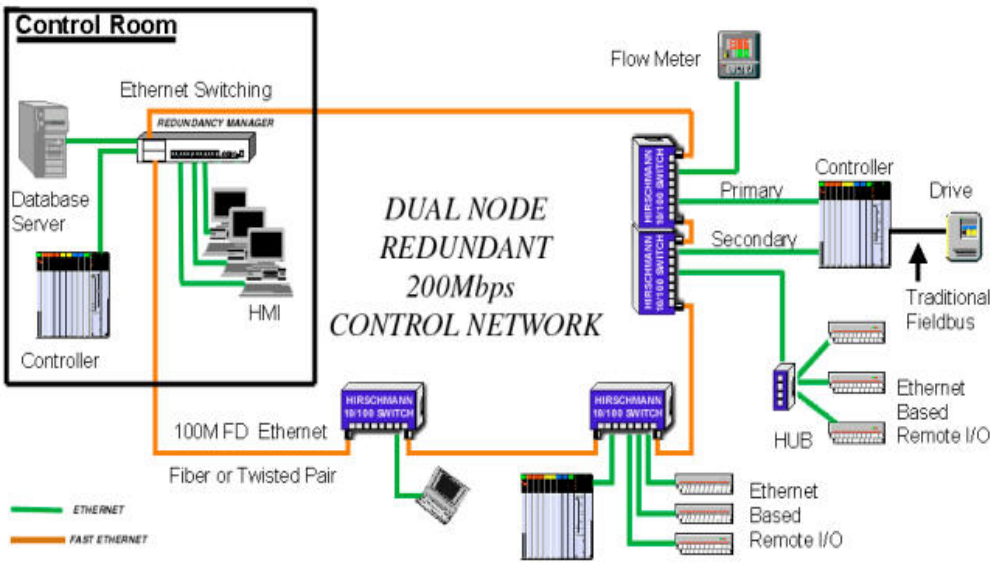


Figure 15.7
Redundant switch ring with dual access to controller
Courtesy: Hirschmann

Ultimately one could protect the system against a total failure by duplicating the switched ring, connecting each port of the dual–homed system to a different ring.

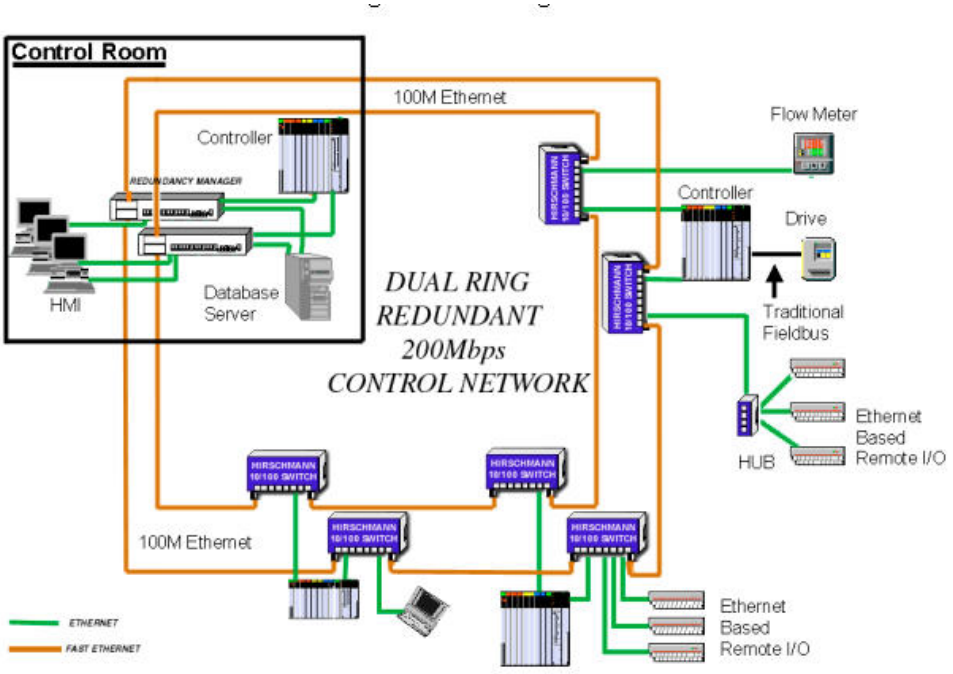


Figure 15.8
Dual redundant switch ring
Courtesy: Hirschmann

Other factors supporting a high degree of resilience would include hot swappable switches and NICs, dual redundant power supplies and on–line diagnostic software.